

6.3 List slicing ^

Settings

Beta features

slice notation

A programmer can use slice notation to read multiple elements from a list that contains only the desired elements.

Figure 6.3.1: List slice notation.

```
boston_bruins = ["Tyler", "Zdeno", "Patrice"]  
print(f"Elements 0 and 1: {boston_bruins[0:2]}")  
print(f"Elements 1 and 2: {boston_bruins[1:3]}")
```

Elements 0 and 1:
Elements 1 and 2:

PARTICIPATION ACTIVITY

6.3.1: List slicing.

Start

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2x speed

```
my_list = ["practicing", "with", "list", "slicing"]  
print(my_list[0:3])  
print(my_list[1:2])
```

['practicing', 'with', 'list']

my_list

0	1	2
"practicing"	"with"	"list"



['with']



Start here



End

Captions ^

1. The list object is created.
2. The list is sliced from 0 to 3, and then printed out.
3. The list is sliced from 1 up to 2.

Figure 6.3.2: List slicing: Using negative indices.

```
election_years = [1992, 1996, 2000, 2004, 2008]
print(election_years[0:-1]) # Every year except the last
print(election_years[0:-3]) # Every year except the last three
print(election_years[-3:-1]) # The third and second to last years
```

stride

An optional component of slice notation is the stride, which indicates how many items are skipped between extracted items in the source list.

Table 6.3.1: Common list slicing operations.

Operation	Description	Example
<code>my_list[start:end]</code>	Get a list from start to end (minus 1).	<code>my_list = [5, 6, 7, 8, 9]</code> <code>print(my_list[1:4])</code>
<code>my_list[start:end:stride]</code>	Get a list of every stride element from start to end (minus 1).	<code>my_list = [5, 6, 7, 8, 9]</code> <code>print(my_list[1:4:2])</code>
<code>my_list[start:]</code>	Get a list from start to end of the list.	<code>my_list = [5, 6, 7, 8, 9]</code> <code>print(my_list[2:])</code>
<code>my_list[:end]</code>	Get a list from beginning of the list to the end (minus 1).	<code>my_list = [5, 6, 7, 8, 9]</code> <code>print(my_list[:4])</code>
<code>my_list[:]</code>	Get a copy of the list.	<code>my_list = [5, 6, 7, 8, 9]</code> <code>print(my_list[:])</code>

CHALLENGE
ACTIVITY

6.3.1: Enter the output of the sliced list.

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Start

Type the program's output

```
my_list = [13, 14, 15, 16, 17, 18, 19]
new_list = my_list[0:2]
print(new_list)
```

[13, 14]

1234

Check

Next

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ACTIVITY**

6.3.2: List slicing.

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Start

List `locations_list` is read from input. Assign `selected_locations` with a slice of `locations_list` from excluding the last two elements.

[Click here for example](#) ▼

```
1 locations_list = input().split()
2
3 """ Your code goes here """
4
5 print(f"Original list of locations: {locations_list}")
6 print(f"Selected list of locations: {selected_locations}")
```

1

2

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